← Back **Week 4 quiz**
Graded Assignment • 20 min

Due Mar 16, 11:59 PM CET

1. With a given telescope we are able to just detect a very faint galaxy by integrating light on a photographic plate for 60 seconds. Using the information from this week's videos, which of the following improvements will allow us to detect another galaxy that is around ten times fainter? 1 point

- ○ Use a telescope with a primary mirror which has half the diameter of the original mirror
- ◉ Replace the photographic plate with a CCD and integrate for the same amount of time
- ○ Use the same telescope but integrate for twice as long
- ○ Use a telescope with a primary mirror which has twice the diameter of the original mirror

2. Why do we need infrared (IR) telescopes for a better understanding of the Universe? [tick all that apply] 1 point

- ☐ Infrared telescopes are cheaper so we can build more of them
- ☐ Infrared detectors are sensitive to light emitted from material entering black holes
- ☑ Infrared detectors reveal light emitted by colder stars in our Milky Way galaxy
- ☑ The most distant galaxies are visible in the infrared and not at optical (visible) wavelengths

3. An infrared photon in the K band has wavelength 2 microns ($\lambda = 2 \times 10^{-6}$ m), and an optical photon in the G band has wavelength 500nm ($\lambda = 5 \times 10^{-7}$ m). Which of the following is true: 1 point

- ○ The IR photon has 16 times the energy of the optical one
- ○ The IR photon has 1/16 the energy of the optical one
- ◉ The IR photon has 1/4 the energy of the optical one
- ○ The IR photon has 4 times the energy of the optical one
- ○ Both photons have equal energy

4. Galaxies that are far away from us are observed to be more red than nearby galaxies because (pick one option) 1 point

- ◉ The Universe is expanding and hence all the galaxies are moving away from us. Their light experiences lots of little Doppler effect shifts as it travels towards us across the expanding Universe. The further away the galaxy is, the more little doppler shifts its light will experience, and the further its light is shifted towards the red.
- ○ The further away a galaxy is, the further back in time we're seeing it, and hence the younger it is. Young galaxies are made of red stars and so the galaxy appears to be red.
- ○ The Universe is expanding. As light travels towards us across the expanding universe, space expands and stretches the light, so it's wavelength increases and the galaxy looks red.
- ○ As light from a distant galaxy travels across the Universe, it slowly loses energy, like a car running out of gas. When the light reaches Earth the lower energy light looks red.

5. Considering the same two photons from the question 3, and given that the optical one has energy 2.5eV, which one of these statements is true? 1 point

- ○ Neither photon would be detected by a silicon photocell, but both would be detected by a MerCadTel photocell
- ○ Only the IR photon would be detected by a silicon photocell
- ◉ Only the optical photon would be detected by a silicon photocell
- ○ Both the optical and the IR photon would be detected by a silicon photocell